

24 | OVERVIEW



Evaluating Competing Claims on Ice Sheet Melting and Sea Level Rise

Abstract

In 2015, a study led by former NASA scientist James Hansen argued that Greenland and Antarctica could melt fast enough to raise sea levels by several meters this century. Climate scientist Kevin Trenberth found the study provocative and intriguing but built on too many assumptions. This lesson uses their disagreement to explore how scientists evaluate bold claims, models, and evidence.

Introduction

Sea level rise threatens coastal cities around the world, so predictions of how fast ice sheets will melt matter enormously. The Hansen study, published online in *Atmospheric Chemistry and Physics*, concluded that ice sheets are vulnerable to non-linear disintegration: rapid melting once temperatures rise past a certain point. It argued that multi-meter sea level rise would become practically unavoidable if emissions keep growing, with possible social disruption, forced migration, and economic collapse. These conclusions differed from the IPCC's 2013 assessment. The paper drew wide attention from scientists and the press. Trenberth agreed that emissions must be cut immediately, but questioned whether the evidence supported such dramatic projections.

How Do Scientists Study This?

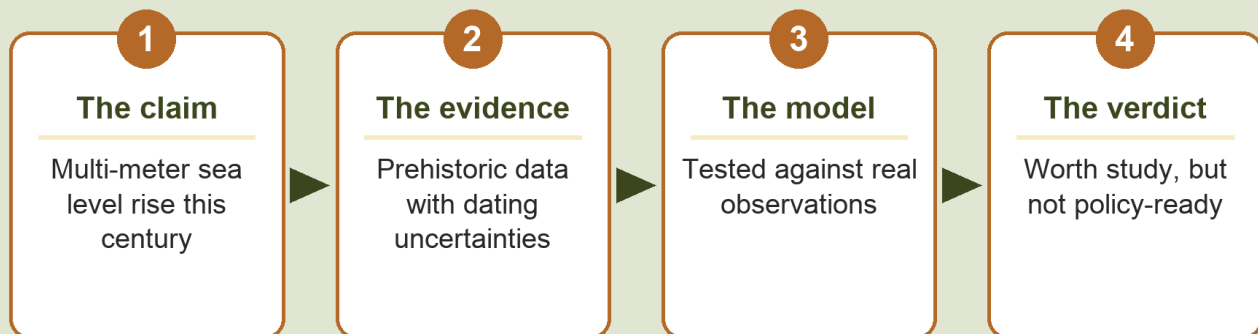


FROM THE ARTICLE

Florida at night from space. Changes in sea level rise projections matter for coastal areas. Image: NASA.

The Hansen study relied on two main kinds of evidence. The first was paleoclimate, or prehistoric, data: clues from the distant past about how ice sheets and sea levels changed. Such data can be controversial because of uncertainties in dating and in converting proxy records into physical quantities. The second was a climate model used for experiments that added large amounts of very cold fresh water at high latitudes to simulate melting ice. Trenberth evaluated these methods the way scientists review any study: checking whether the model reproduces real-world observations, whether key processes like clouds and ocean salinity are simulated well, and whether the conclusions follow from the evidence.

How scientists evaluate a bold claim



Study the Process

Examine the evidence

Check the quality of prehistoric data, including dating and proxy uncertainties.

Test the model

Compare model results with observations of clouds, salinity, and natural variability.

Weigh the conclusions

Ask whether the claims follow from the evidence or stretch beyond it.

What Did They Discover?

Trenberth concluded that the study raised good questions worthy of further research but was rife with speculation and huge extrapolations based on flimsy evidence. The model did not show comparisons with observations, and some errors appeared large. It did not account for short-term natural variability such as El Niño or Pacific decadal variations, which dominate year-to-year and decade-to-decade changes. It simulated sea surface salinity poorly, and like other models, let far too much sunshine reach the surface over the southern oceans. The freshwater experiments did not seem realistic. The authors noted model limitations but did not keep them in mind when drawing conclusions. In his view, the paper was a tour de force, but not a basis for setting policy.

The debate at a glance

Multi-m

sea level rise claimed possible this century

2013

IPCC assessment the study disagreed with

2

ice sheets at issue: Greenland and Antarctica

Trenberth strongly agreed that emissions must be reduced immediately, but argued the study went out of its way to make the case, stretching credibility.

24 | TAKEAWAYS

Takeaways

Bold claims need strong evidence

Dramatic predictions must be backed by reliable data and tested models.

Models must match reality

A model's results are only trustworthy if it reproduces real-world observations.

Scientists can disagree

Experts may share goals but still challenge each other's evidence and methods.

Key Terms

Ice sheet

A huge mass of glacial ice covering land, such as those on Greenland and Antarctica.

Paleoclimate data

Evidence about past climates from sources such as ice cores, sediments, and fossils.

Proxy data

Indirect measurements, like tree rings or chemical signals, used to estimate past conditions.

Non-linear change

Change that does not happen at a steady rate but can speed up suddenly after a threshold is crossed.

Extrapolation

Extending a trend or result beyond the range of the available data, which increases uncertainty.

24 | ACTIVITY

Activity

Peer Review Panel

Students act as reviewers evaluating a bold scientific claim using a simple checklist.

Materials: Peer review checklist; Summary cards of the Hansen study and Trenberth's critique; Chart paper

1. Read the summary of the Hansen study and list its main claims.
2. Use the checklist: Is the data reliable? Does the model match observations? Are limitations addressed? Do conclusions follow from evidence?
3. Score each checklist item and note evidence from Trenberth's critique.
4. Identify one strength of the study that makes it worth further research.
5. Write a short reviewer's recommendation: accept, revise, or reject for policy use, and explain why.

References

Trenberth, K. (2015). Study predicts multi-meter sea level rise this century, but not everyone agrees. The Conversation. <https://doi.org/10.64628/AAI.twsmjdaqm>

Hansen, J., et al. (2015). Atmospheric Chemistry and Physics (discussion paper), as described in the article above.